09E — Deep-Dive Feature Audit: Combat Engine & Tactical Simulation (Part 5 of 6)

Fight Modes & Stakes Lifecycle: Impossible Raid `CORE_BREACHED` AI, Self-Defense Grudge Jail Bug, `8`-Layer Cheat-Death & Spar Heart-Demon Leakage, Unbracketed Tournaments & The `15`-Day Heavenly Tribulation Clock Bug

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` in Google Drive `Js2 new`)
- **Target Audience**: Autonomous AI Developers (**GPT** & **Gemini**)
- **Core Game Philosophy**: **100% Emergent, Context-Appropriate Combat Stakes**. Friendly spars must never detonate blood-taboo bombs or inflict `+10` Heart Demons; victims defending their lives in grudge ambushes must not be jailed as murderers; raiders must storm the Main Hall if defenders flee to map corners; and Stage-9 Patriarchs must autonomously brave Heavenly Ascension!
- **Master Roadmap for Category 09 (`09A–09F`)**:
  - `09A`: Core Damage Formula (`dmg = hpMax(d) * frac`), 12-Stat Split-Brain, Elemental Matchups & Behavioral Mastery Traits
  - `09B`: Spatial Battlefield Geometry, `NPC` vs `foe` Movement Asymmetry, `24`-Tile `ROW_BODYBLOCK` Bug, Interception & AoE Selectors
  - `09C`: Action Economy, Charged/Channeling Multiplier Loss, `statusTick` DoT Loops, Shield Overwrite, Acupoints & Boss Anatomy Breaker
  - `09D`: Multi-Entity Units — Summons, Shadow Clones, Constructs, Pet Assists, Gate Beasts, Weapon Avatars, Phantoms & Party Synergies
  - **`09E` (This Document)**: **Fight Modes & Stakes — Raids, Grudge/Leadership Duels, Spars, Tournaments, Cheat-Death Stacking, Mercy vs `VOW_NO_KILL` & Dual Heavenly Tribulations (`combat.js`, `sim.js`, `cultivation.js`, `sect.js`)**
  - `09F`: Grand Finale — Holistic Overview of the Entire Combat Ecosystem

---

1. [CMB-TRIB-01] Critical Time-Scale & Split-System Bug in Heavenly Tribulations (js/sim.js:465 vs test/v530:97–128 & test/v211): A 15-Tick Lightning Tribulation Takes 15 Full Days & Stage-9 Elders Never Ascend Autonomously!

1. **CULT.TRIBULATION.tickAll(W) Wired Into SIM.dailyUpdate Instead of SIM.tick (js/sim.js:465)!**:
   - Inspect v530_heavenly_tribulation_and_skill_vfx_suite.js:97–128: CULT.TRIBULATION is a **15-tick** (15 simulation minutes) 3-phase lightning breakthrough (4 ticks GATHERING $\to$ 7 ticks STRIKING $\to$ 1 tick ASCENSION/BACKFIRE $\to$ 3 ticks cleanup).
   - **Yet in js/sim.js, CULT.TRIBULATION.tickAll(W) is called ONLY at line 465 inside SIM.dailyUpdate (once every 1,440 ticks at midnight!)—NEVER inside SIM.tick (lines 322–345)!**
   - **Consequence**: In live gameplay, a 15-tick lightning tribulation freezes the sky purple for **15 full days (21,600 minutes — 1 lightning bolt every 58 hours!)**!
2. **Two Disconnected, Player-Button-Locked Tribulation Systems (CULT.TRIBULATION vs SECT.beginCelestialTribulation)**:
   - CULT.TRIBULATION.start (v5.3.0) is triggered **only** by UI.doAction('start-tribulation-ritual') (0% connection to actual major-realm A.breakthrough in ai.js!), AND ignores EFX_TRIBULATION_SHIELD (05C) and 護法 Dharma-Protector arts (03F)!
   - Meanwhile, Stage-9 Immortal Ascension (SECT.beginCelestialTribulation in v211, costing 500 stones — **also verify those 500 stones credit W.world.resourcePool.stones!**) is **also 100% UI-button locked** (0 Stage-9 Patriarchs ever attempt Immortal Ascension on their own in 300-year runs!).

- **Exact Fix**:
  - Move CULT.TRIBULATION.tickAll(W) into SIM.tick (running only when W.sect.activeTribulations?.length > 0, O(1) zero overhead when idle!), trigger CULT.TRIBULATION.start automatically whenever an NPC attempts a major stage >= 3 breakthrough (Golden Core+), apply EFX_TRIBULATION_SHIELD (-28%..32% bolt damage), and let Stage-9 (equivalentStage === 9) Elders with high martialPeak/dao values (and appointed successor if Master!) autonomously initiate SECT.beginCelestialTribulation!

---

2. [CMB-RAID-02] Raid Inner-Sanctum Breach (CORE_BREACHED) Is Blocked by Omniscient Map-Wide Targeting & Kiting Auto-Wins via REPELLED_TIMEOUT (js/combat.js:704–755)!

Inspect C.foeStep (lines 704–714) alongside C.tick (lines 739–755):

```JavaScript
const t = eng.target;
if (!t) {
  if (C.sideOf(f, u.id) === 'B' && f.kind === 'raid') { moveToward(W, u, coreX,
coreY, false, f); ... }
  return;
}
u.reachedCore = false;
```

1. **Why Raiders Never Storm main_hall While 1 Defender Hides Across the Map**:
    - C.enemies(W, f, u.id) returns **every** living non-fled defender in f.A across the entire 240×192 map with **no detection radius**!
    - As long as even **1 Outer Disciple stands 160 tiles away at the North Mine**, eng.target (t) is non-null—so raiders at the breached South Gate **abandon marching on main_hall**, set u.reachedCore = false, and jog across the mountain chasing 1 novice!
2. **900-Tick Kiting Auto-Win**:
    - If 1 fast QINGGONG disciple kites raiders past elapsedFight > 900 (line 754), f.winner = 'A'; outcome = 'REPELLED_TIMEOUT' grants a full defensive victory even if the raiders demolished the Gate and stood outside an undefended Main Hall!

- **Exact Fix**: In C.foeStep for f.kind === 'raid' (side === 'B'), if the nearest living defender is > 16.0 tiles away (no defender contesting the approach!), raiders **advance directly on main_hall (coreX, coreY)**—forcing defenders to actually hold the Gate / Plaza rather than kiting at the map border!

---

3. [CMB-STAKES-03] Friendly-Spar Taboo Explosions & +10 Heart-Demon Leak (js/combat.js:382, 464, 479), Self-Defense Grudge Jail Bug (line 795), & Unbracketed Tournaments (line 804)!

1. **Non-Lethal Spars & Tournaments Trigger Blood-Taboo Explosions & Inflict `+10 heartDemon` (`lines 382 & 479–495`)!**:
   - **Bug A (`line 479` vs `494`)**: `KW_LAST_BREATH` (`intrinsic to all DEMON, ARENA_GLORY, and PRIMORDIAL_SHIFTER disciples!`) triggers at `if (hp <= 0 && !d.isSummon)` on line `479`—**15 lines BEFORE `if (nonlethal) { d.yield = true; }` (`line 494`)**—inflicting **`+10 heartDemon` on friendly spar losers every single match**!
   - **Bug B (`line 382`)**: `TABOO_DESPERATION_NOVA` (`40% MaxHP` blood-void explosion) does **not** check `f.serious`, detonating lethal taboo energy on junior sectmates during friendly spars!
   - **Bug C (`Uncapped 8-Layer Cheat-Death Stacking`)**: Cap total fatal-save / cheat-death triggers (`Cicada Molt`, `Substitution`, `Desperation Nova`, `Phoenix Pet`, `Mech Decoy`, `Karma Echo`, `nirvanaGrace`, `phoenixCovenant`, `KW_LAST_BREATH`) to **at most `2` saves per unit per fight** (`fTac.savesUsedCount[d.id] < 2`), and guard `KW_LAST_BREATH` + `TABOO_DESPERATION_NOVA` behind `if (f && f.serious)`!
2. **Innocent Self-Defense Victims Jailed in `grudge` Duels & Forced `VOW_NO_KILL` Violations (`lines 633, 795`)**:
   - In `C.duelEnd` (`line 795`), if aggressor `f.A[0]` starts a lethal grudge fight and target `f.B[0]` kills `f.A[0]` in **self-defense**, `w = f.B[0]` is **jailed for 3 days** (`w.jail = W.t + 3 * 1440`)! Only jail `w` if `w.id === f.A[0]` (the aggressor) or if `w` executed a yielding opponent!
   - For cultivators upholding **`VOW_NO_KILL` / `VEGETARIAN_PACIFIST` (`daoOath`) or `VOW_NO_KILL` (`voluntaryVows`)**, convert lethal blows (`hp <= 0`) against humanoids/non-fiends into **Merciful Subdual (`hp = 1; d.yield = true; fx: '🙏 ออมมือสยบ!'`)** unless enraged (`heartDemon >= 75`)—so Buddhist monks don't robotically murder bandits and break their own precepts! Similarly, block `50%` flee rolls at line `505` for **`VOW_NO_RETREAT`** holders (`unless courage breaks under extreme terror >= 90`)!
3. **Unseeded Sect Tournament (`C.tournamentTick`, `lines 804–845`)**:
   - Group `W.sect.tournament` into tiered brackets (`or seed pairings by rank/stage`) so Stage-1 Outer Novices aren't fed to Stage-6 Elders in Round 1, and use `SECT.bindMentorship(W, m, win)` at line `843` so tournament-recruited apprentices get `mentorId` + `apprenticeIds`!

---

4. Mandatory Verification Suite Additions (09E)

1. **15-Minute `CULT.TRIBULATION` & Non-Lethal Spar Purity Test**: Trigger `CULT.TRIBULATION.start`, run `15` calls of `SIM.tick(W)`, and assert the tribulation resolves within `15` minutes (not `15` days!); run a non-lethal `spar` between two `DEMON` (`KW_LAST_BREATH`) disciples to defeat and assert `+0 heartDemon` gained!
2. **Raid Undefended-Gate Advance Test**: Spawn a `raid` at `(118, 132)` with the sole defender standing idle at `(30, 20)`; assert raiders advance to `main_hall` and increment `breachTicks` rather than chasing across the mountain!
3. **Self-Defense Grudge & `VOW_NO_KILL` Mercy Test**: Have aggressor A attack defender B (`VOW_NO_KILL`) in a `serious: true` grudge; assert B subdues A at `1` HP without breaking `VOW_NO_KILL` or going to `jail`.